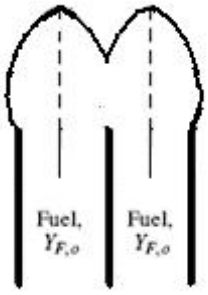
1　分析题（Answer the following questions briefly）（**7题，共44分**）

1.1　(6 分) 对于层流扩散火焰，当两个同样流量的射流喷口紧密地排列在一起进行燃烧时(如下图)，此时每个喷口仍保持原来的流量，那么请问与单个喷口相比，火焰长度如何变化，并给出原因。

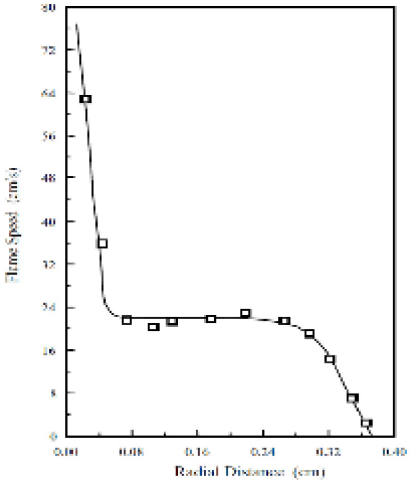
As for laminar jet diffusion flame, if two jet nozzle are closely lined together but keep the same operational parameters, what is changes of flame height of each jet flame in contrast to single jet one, and give the explanation. (6 分)



答案提要：火焰变长，两个临近的地方上方氧气的含量降低，更难到达当量比等于 1 的面

1.2　(8 分) 如图测定的是本身灯的真实火焰传播传度的随径向分布值，本身灯喷口的直径为 8mm。试分析出现这一分布的原因，并由此图绘出火焰的形状。

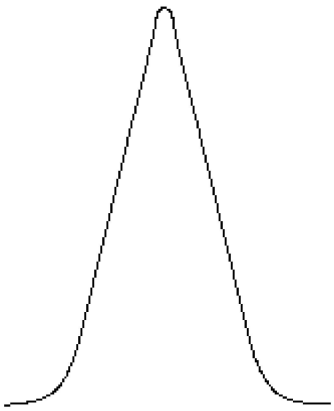
The figure below gives the true measurement of the radial distribution of flam speed by a Bunsen burner, please analyze the reason that causes this distribution, and try to draw the flame shape of Bunsen burner according to this experimental result.



答案提要：

在边缘地区由于散热使得温度降低，同时由于外界低温二次风的混入，使得火焰速度降低；而在靠近中心的地方温度较高，未然气体的温度也有所提高，导致了火焰速度的迅速提高；而在中间区域位置没有二次风的影响，未燃气体温度也基本不变，所以火焰速度基本保持不变。

另外，气流的速度沿着径向的分布是在靠近管壁的地方由于边界层的存在接近于零，而在其他地方的速度基本保持不变。因此，火焰的形状大致为：



1.3 (6 分)碳烟通常产生于非预混火焰，请结合课本知识简单阐述碳烟形成机理(4 分)，如果煤燃烧过程也会产生碳烟，试根据煤燃料的特点分析产生这些碳烟的具体原因 (2 分)。

Please describe the forming mechanism of soot in diffusion flames according to your knowledge. We know that coal often produce soot during its combustion. What is the original source of soot formation from coal fuels? Please try to explain it.

答案提要：

机理：前体物形成；颗粒形成；颗粒长大和团聚；氧化

挥发份析出和燃烧会有大量的碳氢化合物例如多环芳烃的形成，这样就比较容易形成碳烟的前体物，形成初始颗粒，然后逐渐的长大和聚合，而在挥发份火焰的内部区域由于氧气无通过扩散方式进入，导致内部处于缺氧的状态，只有挥发份中的少量含氧的基团对碳烟颗粒进行氧化，这样就会在煤燃烧的初期形成碳烟。

1.4 (6 分) 甲烷在三种不同混合物气氛下进行燃烧：(1) O2-N2、(2) O2-Ar、(3) O2-CO2。三种气氛混合物的比例都是 21/79，试比较当量比为 1 时三种气氛下甲烷的火焰速度(S_L)大小，并分析原因。**提示：从惰性介质不同引起火焰温度或其它性质等变化角度来看。**

The methane can burn in three different mixtures: (1) O2-N2、(2) O2-Ar、(3)O2-CO2. The mixing ratios are all 21/79 for the oxidizer. Please try to compare the flame speeds in these three mixtures as well as give a sequence. Then to give the reason for that.

答案提要：单原子，双原子，三元子，定压比热焓依次增加，所以火焰的温度会依次降低，火焰速度也会依次降低。

1.5 (7分) 一般家用的炉灶用钢材作为燃烧器的管子的材料（其热导率为100W/m-K），假设燃料为甲烷，原燃烧器共有30个喷口，原燃烧器的喷口布置在直径为100mm的圆周上。有人提出用陶瓷（热导率约30W/m-K）来作为管子材料。试分析会出现什么问题，试定性分析需要作何调整以确保燃烧器可用。

The cooktop burner uses the steel as the material of burner pipe (heat conductivity is 100W/m-K), and some people attempts to use ceramic as the material (heat conductivity is 30W/m-K). Please try to analyze what will occur. And try to think out the modifications of the

burner for the formal utilization.

答案提要：将管子材料由钢材换为陶瓷之后，最主要的问题就是防止回火，因为材料的导热系数发生了很大的变化，而且气体和陶瓷管之间的换热系数也发生了变化。可以采用多根燃料管减小管道的直径的方法来防止回火。

1.6 (5 分) 液滴和气流相对运动越快，则液体燃烧速率如何变化，请给出简要的分析。

If the velocity between droplet and ambience fluid become higher, what will the burning rate changes? And give your reason.

答案提要：相对运动越快，则 Nu 数越大，根据公式 10.84a，液体燃烧速率会增加。

1.7 (6 分) 试解释守恒标量的定义，除了混合物分数和混合物绝对焓，请根据自己对组分守恒方程和能量方程的理解，写出一种新的守恒标量。

Please try to list out the definition of the conserved scalar, beside the mixture fraction and mixture enthalpy, can you give another new conserved scalar?

答案提要：流场内满足守恒的任一标量，在守恒方程中没有生成项和消耗项。例如：某一个元素的质量分数。

2 (12分)在碳颗粒燃烧的**双膜**模型中，有两个重要的中间公式 $\frac{Y_{CO_2,f}}{\nu_s}=\frac{Y_{O_2,\infty}}{\nu_s-1+Y_{O_2,\infty}}$ 和 $\frac{\dot{m}_C}{r_s}=4\pi\rho D\ln\left[\left(\frac{1+Y_{CO_2,f}/\nu_s}{1+Y_{CO_2,s}/\nu_s}\right)\left(\frac{1}{1-Y_{CO_2,f}/\nu_s}\right)\right]$，试简述其推导思路和过程(6分)？根据这个式子，可否写出其**碳表面**所对应的CO_2扩散阻力R_{diff} (3分)？如果采取纯氧条件下富氧燃烧方式，试问扩散控制条件下，和空气燃烧相比**碳表面**的扩散阻力的变化，即R_{diff}(纯氧)/R_{diff}(空气)比值(3分)。**提示：扩散阻力定义是质量分数势差和燃烧速率之比。**

In the two film model, please briefly derive the two important equation of $\frac{Y_{CO_2,f}}{\nu_s}=\frac{Y_{O_2,\infty}}{\nu_s-1+Y_{O_2,\infty}}$ and $\frac{\dot{m}_C}{r_s}=4\pi\rho D\ln\left[\left(\frac{1+Y_{CO_2,f}/\nu_s}{1+Y_{CO_2,s}/\nu_s}\right)\left(\frac{1}{1-Y_{CO_2,f}/\nu_s}\right)\right]$, . Then write out the diffusion resistance of CO2 in the inner zone, R_{diff}. If the coal are burning in the pure oxgen, please evaluate the ratio of R_{diff}(Oxgen)/R_{diff}(Air).

答案提要：

根据书上14.46-14.48即可以推导出来质量分数的式子 $\frac{Y_{CO_2,f}}{\nu_s}=\frac{Y_{O_2,\infty}}{\nu_s-1+Y_{O_2,\infty}}$；根据14.45和14.46就可以推导出燃烧速率的式子 $\frac{\dot{m}_C}{r_s}=4\pi\rho D\ln\left[\left(\frac{1+Y_{CO_2,f}/\nu_s}{1+Y_{CO_2,s}/\nu_s}\right)\left(\frac{1}{1-Y_{CO_2,f}/\nu_s}\right)\right]$，。

由此推出：

$$R_{diff}=\frac{Y_{CO_2,f}-Y_{CO_2,s}}{\dot{m}_C}=\frac{\frac{\nu_s Y_{O_2,\infty}}{\nu_s-1+Y_{O_2,\infty}}-Y_{CO_2,s}}{4\pi r_s\rho D\ln\left[\left(\frac{1+Y_{CO_2,f}/\nu_s}{1+Y_{CO_2,s}/\nu_s}\right)\left(\frac{1}{1-Y_{CO_2,f}/\nu_s}\right)\right]}$$

$$R_{diff}=\frac{\frac{\nu_s Y_{O_2,\infty}}{\nu_s-1+Y_{O_2,\infty}}-Y_{CO_2,s}}{4\pi r_s\rho D\ln\left[\left(\frac{\nu_s-1+2Y_{O_2,\infty}}{1+Y_{CO_2,s}/\nu_s}\right)\left(\frac{1}{\nu_s-1}\right)\right]}$$

在扩散控制的条件下，在纯氧燃烧的工况下， $Y_{CO_2,s}=0, Y_{O_2,\infty}=1$，

$$R_{diff}=\frac{\frac{3.664}{3.664-1+1}-0}{4\pi r_s\rho D\ln\left[\left(\frac{3.664-1+2}{1+0}\right)\left(\frac{1}{3.664-1}\right)\right]}=\frac{1}{4\pi r_s\rho D}\frac{1}{\ln(1.751)}=\frac{1.786}{4\pi r_s\rho D}$$

在空气工况下，$Y_{CO_2,s}=0, Y_{O_2,\infty}=0.233$，

$$R_{diff}=\frac{\frac{3.664\times0.233}{3.664-1+0.233}-0}{4\pi r_s\rho D\ln\left[\left(\frac{3.664-1+2\times0.233}{1+0}\right)\left(\frac{1}{3.664-1}\right)\right]}=\frac{1.828}{4\pi r_s\rho D}$$

R_{diff}(纯氧)/R_{diff}(空气)=0.977

3 (12分)用于加热炉的燃烧器一般采用扩散火焰，试进行一个加热炉内燃烧器的设计，采用天然气作为燃料（以甲烷计），总功率为100kW，假设燃烧器的火焰处于层流，要求火焰的长度大约在3米左右，如果采用单一燃烧器喷口并用纯扩散火焰，长度大大超过要求。请问你如何设计你的燃烧器以满足这一要求，需要定量计算与分析。

The heater furnace often adopts diffusion flames. Here is a design of burner with methane as the fuel. The total heating power is 100 kW. If there is only one jet burner, the flame height may exceed the limit of 3 meter. How do you solve this problem, please give the calculation and analysis.

答案提要：

设未燃气体的温度为298K，总燃气流量为：

$$\dot{m}=\frac{100\text{kW}}{50016\text{kJ/kg}}=1.999\times10^{-3}\text{kg/s}，\quad Q=\frac{\dot{m}}{\rho}=\frac{\dot{m}}{\frac{PMW}{R_u T}}=\frac{1.999\times10^{-3}}{\frac{101325\times16}{8315\times298}}=0.00306\text{m}^3/\text{s}$$

对于甲烷，S=9.52，取$T_\infty = T_F = 298K$，若是采用单一燃烧器喷口，则火焰长度为：

$$L_f = 1330\frac{Q_F(T_\infty / T_F)}{\ln(1+1/S)} = 40.75\,\text{m}$$

所以，为了使得火焰的长度减为3m，可以采用多个喷嘴或是加入一次风的方法来降低火焰长度。以采用多个喷嘴的方法为例，为使火焰长度减为3m，总的流量与单根管的流量之比为：

$Q_F / Q'_F = 40.75/3 = 13.58$，

所以，可以采用14根管。这样，火焰的长度为：$L_f' = 40.75/14 = 2.91\text{m}$

4 (14分) 直径200μm的碳颗粒在静止空气($Y_{O_2,\infty} = 0.233$)中燃烧，**燃烧模式由扩散阻力控制，**其中化学动力学方程可以表示为$\dot{m}_C = 4\pi r_s^2 k_p MW_C P_{O_2,s}^{0.5}$，其中$k_p = 4\times10^{-2}\exp(-10072/T_s)$，其他条件：压力为101325Pa；碳表面温度$T_S$=1500K；；$\mathcal{D}(293K) = 1.6\cdot10^{-5}$ [m^2/s]。下面计算均基于单膜模型来开展。

(a)如果气相混合物的摩尔质量假设为MW_{mix}=32 kg/kmol，MW_C=12 kg/kmol请计算出碳燃烧速率$\dot{m}_C$(5分)；If the molecular weight of gas mixture is MW_{mix}=32 kg/kmol, MW_C=12 kg /kmol，please determine the carbon burning rate $\dot{m}_C$(5分);

(b) 确定碳表面处的O_2的质量分数$Y_{O_2,s}$ (4分)；

Please determine the mass fraction of O_2 at carbon surface, $Y_{O_2,s}$. (4分). Clue: this work can be started by derive the diffusion equation of CO_2 in the single-film model and compare it with that of O_2.

(c) 试计算假定扩散控制条件下，碳表面处的CO_2的质量分数$Y_{CO_2,s}$，提示：从CO_2的扩散方程出发建单膜模型并和O_2的对比(5分)；
Under the diffusion-controlled regime, please determine the mass fraction of CO_2 at carbon surface, $Y_{CO_2,s}$. (5分). Clue: this work can be started by derive the diffusion equation of CO_2 in the single-film model and compare it with that of O_2.

答案提要：

a) 扩散控制，$Y_{O_2,s} = 0$，取1500K为定性温度，

计算得$\rho = 0.260\text{kg/m}^3$，$D = 1.85\times10^{-4}\text{m/s}$，

$$R_{diff}=\frac{\nu_1+Y_{O_2,s}}{\rho D 4\pi r_s}=\frac{2.664+0}{4\times 3.14\times 0.260\times 1.85\times 10^{-4}\times 100\times 10^{-6}}=4.41\times 10^{7}\,\mathrm{s/kg}\,,$$

$$\dot{m}_C=\frac{Y_{O_2,\infty}}{R_{diff}}=\frac{0.233}{4.41\times 10^{7}}\,\mathrm{kg/s}=5.283\times 10^{-9}\,\mathrm{kg/s}$$

b) 而：

$$\dot{m}_C=4\pi r_s^2 k_p MW_C P_{O_2,s}^{0.5}=4\pi r_s^2 k_p MW_C P^{0.5} Y_{O_2,s}^{0.5}$$

带入数据解得$Y_{O_2,s}=0.0514$，对扩散阻力的影响接近2%，是扩散控制的。

c) 由组分守恒方程有：

$$\dot{m}_{CO_2}=Y_{CO_2}(\dot{m}_{O_2}+\dot{m}_{CO_2})-4\pi r^2\rho D\frac{\mathrm{d}Y_{CO_2}}{\mathrm{d}r}\,,$$

而$\dot{m}_{O_2}+\dot{m}_{CO_2}=\frac{1}{\nu}\dot{m}_{CO_2}$，因此上式变为：

$$\dot{m}_{CO_2}\frac{\mathrm{d}r}{r^2}=-4\pi\rho D\frac{\mathrm{d}Y_{CO_2}}{1-\frac{Y_{CO_2}}{\nu}}\,,$$

从C表面到无穷远处积分可以得到：

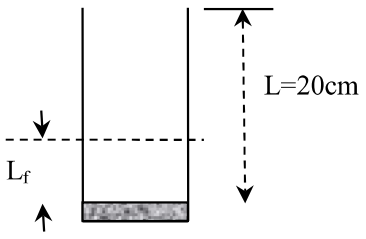
$$\dot{m}_{CO_2}/r_s=4\pi\rho D\nu\ln\frac{\nu}{\nu-Y_{CO_2}}$$

而$\dot{m}_{CO_2}=3.664\,\dot{m}_C=3.664\times 5.283\times 10^{-9}\,\mathrm{kg/s}=1.9\times 10^{-8}\,\mathrm{kg/s}$，带入解得：

$$Y_{CO_2,s}=0.297$$

5 (18 分) 在 1 个器皿中含有一定厚度的液膜，成分辛烷 C_8H_{18}，且保持高度始终低于管口 L＝20cm,在壁面绝热环境下发生了蒸发燃烧，假设燃烧是层流，且满足一维情况，管上方为空气流，液体的温度已经达到液体燃料的沸点温度，外界的温度为 300K.

Consider an adiabatic cylinder containing a thin liquid-fuel film (C_8H_{18}), and the liquid is replenished at a rate to keep the liquid level at a constant height with a L=20cm, as shown in Figure below. We assume that the burning of liquid fuel film is in a laminar, one-dimensional mode. The temperature of liquid is at the boiling temperature and the ambience temperature is 300K.



(1) 根据组分守恒方程，**试证明**液体蒸发燃烧过程如下两个公式（6 分）
According to the species conservation equation in both inner and outer regions, please derive the following two equations:

a. 火焰内区：燃料组分和辛烷燃烧通量的关联式 $\dot{m}_F'' = \frac{\rho D}{L_f}\ln(\frac{1}{1-Y_{F,s}})$

b. 火焰外区：空气组分和辛烷燃烧通量的关联式 $\dot{m}_F'' = \frac{\rho D}{L-L_f}\ln(\frac{\nu+1}{\nu})$

答案提要：

在火焰内区，应用 Fick 定律有：$\dot{m}_F = Y_F(\dot{m}_F + \dot{m}_{Pr}) - \rho D\nabla Y_F$，

而 $\dot{m}_{Pr} = 0$，故有：$(1-Y_F)\dot{m}_F = -\rho D\frac{dY_F}{dl}$，

分离变量由 0 到 L_f 积分有：$\dot{m}_F'' = \frac{\rho D}{L_f}\ln(\frac{1}{1-Y_{F,s}})$

在火焰外区，应用 Fick 定律有：$\dot{m}_{Ox} = Y_{Ox}(\dot{m}_{Ox} + \dot{m}_{Pr}) - \rho D\nabla Y_{Ox}$，

此时 $\dot{m}_{Pr} = (\nu+1)\dot{m}_F$，$\dot{m}_{Ox} = -\nu\dot{m}_F$ 整理有：$(Y_{Ox} - \nu)\dot{m}_F = -\rho D\nabla Y_{Ox}$，

分离变量由 L_f 到 L 积分有：$\dot{m}_F'' = \frac{\rho D}{L-L_f}\ln(\frac{\nu+1}{\nu})$

(2) 对于火焰内区和外区的能量方程，本题目**给定**如下，(4 分)
If the energy conservation equation in both inner and outer regions are given as below,

火焰内区：液滴表面能量方程 $\frac{c_{pg}(T_f - T_{boil})}{h_{fg}} = \exp(m_F''\frac{c_{pg}}{k_g}L_f) - 1$

火焰外区：火焰面上的能量方程 $\frac{c_{pg}}{\Delta h_c}\left[\frac{(T_f - T_{boil})\exp(m_F''\frac{c_{pg}}{k_g}L_f)}{\exp(m_F''\frac{c_{pg}}{k_g}L_f) - 1} - \frac{(T_\infty - T_f)}{\exp[m_F''\frac{c_{pg}}{k_g}(L-L_f)] - 1}\right] = 1$

如果满足 Lewis 数为 1，即 $D = k_g/(\rho c_{pg})$

试证明：液膜的燃烧通量的公式 $m_F'' = \frac{k_g}{c_{pg}L}\ln[1 + \frac{\Delta h_c/\nu + c_{pg}(T_\infty - T_{boil})}{h_{fg}}]$

If Lewis number is 1, e.g., $\rho D = k_g/c_{pg}$, then please derive the equation of liquid burning flux,

$$m_F'' = \frac{k_g}{c_{pg}L}\ln[1+\frac{\Delta h_c / \nu + c_{pg}(T_\infty - T_{boil})}{h_{fg}}]$$

答案提要：

$D = k_g/(\rho c_{pg})$，即：$c_{pg}/k_g = 1/(\rho D)$，

由（1）的证明有：$\dot{m}_F'' = \frac{\rho D}{L_f}\ln(\frac{1}{1-Y_{F,s}})$，$\dot{m}_F'' = \frac{\rho D}{L-L_f}\ln(\frac{\nu+1}{\nu})$

即

$$\exp(\frac{\dot{m}_F'' L_f}{\rho D}) = \frac{1}{1-Y_{F,s}} \qquad (a)$$

$$\exp(\frac{\dot{m}_F''(L-L_f)}{\rho D}) = \frac{\nu+1}{\nu}, \qquad (b)$$

$$\text{而：}\ \frac{c_{pg}(T_f - T_{boil})}{h_{fg}} = \exp(m_F''\frac{c_{pg}}{k_g}L_f) - 1 \qquad (c)$$

将式（c）和（b）代入火焰内区和外区的能量方程有：

$$\frac{C_{pg}}{\square h}[T_f - T_{boil} + \frac{h_{fg}}{C_{pg}} - \nu(T_\infty - T_f)] = 1$$

$$\text{化简有：}\quad (1+\nu)(T_f - T_{boil}) = \frac{\square h}{C_{pg}} - \frac{h_{fg}}{C_{pg}} + \nu(T_\infty - T_{boil}) \qquad (d)$$

而（a）*（b）有：

$$\exp(\frac{\dot{m}_F'' L}{\rho D}) = \frac{\nu+1}{\nu}\frac{1}{1-Y_{F,s}}, \qquad (e)$$

由（a）与（c）式有：

$$\frac{1}{1-Y_{F,s}} = \frac{C_{pg}(T_f - T_{boil})}{h_{fg}} + 1 \qquad (f)$$

由（e）与（f）有：

$$\exp(\frac{\dot{m}_F'' L}{\rho D}) = \frac{\nu+1}{\nu}[\frac{C_{pg}(T_f - T_{boil})}{h_{fg}} + 1] \qquad (g)$$

将（d）代入（g）有：$\exp(\frac{\dot{m}_F'' L}{\rho D}) = 1 + \frac{\Delta h_c / \nu + c_{pg}(T_\infty - T_{boil})}{h_{fg}}$，而$c_{pg}/k_g = 1/(\rho D)$，化简有：

$$m_F'' = \frac{k_g}{c_{pg}L}\ln[1+\frac{\Delta h_c / \nu + c_{pg}(T_\infty - T_{boil})}{h_{fg}}]$$

(3) 根据前面的公式，分别计算火焰离液面的高度$\dot{m}_F''$, L_f,和$Y_{F,s}$（8 分）

According to all the equations above, please predict the $\dot{m}_F''$, L_f , and $Y_{F,s}$

（总的提示：如果证明有困难，可以根据公式进行计算。 Clue: if the derivation is difficult to you, please use the above equations to perform the calculation. ）

相关物性参数：$C_{p,F}(1350K)=4.28\text{kJ/kg-K}$，$k_F(500\text{K})=0.0334\text{W/m-K}$

答案提要：

$T_s=T_{boil}=125.7°\text{C}=398.9\text{K}$，取火焰温度为2300K，所以，定性温度取为：$\overline{T}=\frac{1}{2}(T_s+T_f)=1350K$，

故$C_{p,F}(1350K)=4.28\text{kJ/kg-K}$，$\nu=15.03$

$$\begin{aligned}k_g&=0.4k_F+0.6k_{Ox}\\&=0.4*0.0334*(1350/500)^{1/2}+0.6(86.5*10^{-3})\\&=0.0739W/m\cdot K\end{aligned}$$

所以，

$$\begin{aligned}m_F''&=\frac{k_g}{c_{pg}L}\ln[1+\frac{\Delta h_c/\nu+c_{pg}(T_\infty-T_{boil})}{h_{fg}}]\\&=\frac{0.0739}{4280*0.2}\ln[1+\frac{44791000/15.03+4280(300-398.9)}{300000}]\\&=1.95*10^{-4}\,\text{kg/s}\cdot\text{m}^2\end{aligned}$$

而：

$$\begin{aligned}L_f&=L-\frac{\rho D}{\dot{m}_F''}\ln(\frac{\nu+1}{\nu})\\&=L-\frac{k_g}{C_{pg}\dot{m}_F''}\ln(\frac{\nu+1}{\nu})\\&=0.2-\frac{0.0739}{4280*1.95*10^{-4}}\ln(\frac{15.03+1}{15.03})\\&=0.194\,\text{m}\end{aligned}$$

而$\exp(\frac{\dot{m}_F''L_f}{\rho D})=\frac{1}{1-Y_{F,s}}$，故有：